

LAB ASSIGNMENT-3

DDL Questions (Data Definition Language)

1. Create the TEACHER table with the following attributes:

(Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).

```
CREATE TABLE TEACHER (  
    Name VARCHAR(50),  
    DeptNo INT,  
    Date_of_Joining DATE,  
    DeptName VARCHAR(50),  
    Location VARCHAR(50),  
    Salary INT  
);
```

OUTPUT:

	Field	Type	Null	Key	Default	Extra
►	Name	varchar(50)	YES		NULL	
	DeptNo	int	YES		NULL	
	Date_of_Joining	date	YES		NULL	
	DeptName	varchar(50)	YES		NULL	
	Location	varchar(50)	YES		NULL	

2. Create the SALES table with the following attributes:

(Sales_No, Salesname, Branch, Salesamount, DOB).

```
CREATE TABLE SALES (  
    Sales_No INT PRIMARY KEY,  
    Salesname VARCHAR(50),  
    Branch VARCHAR(50),  
    Salesamount INT,  
    DOB DATE  
);
```

OUTPUT:

	Field	Type	Null	Key	Default	Extra
►	Sales_No	int	NO	PRI	NULL	
	Salesname	varchar(50)	YES		NULL	
	Branch	varchar(50)	YES		NULL	
	Salesamount	int	YES		NULL	
	DOB	date	YES		NULL	

3. Create the DEPARTMENTS table with the following attributes:(dept_no, dept_name, dept_location).

```
CREATE TABLE DEPARTMENTS (  
    dept_no INT PRIMARY KEY,  
    dept_name VARCHAR(50),  
    dept_location VARCHAR(50)  
) ;
```

OUTPUT:

	Field	Type	Null	Key	Default	Extra
►	dept_no	int	NO	PRI	NULL	
	dept_name	varchar(50)	YES		NULL	
	dept_location	varchar(50)	YES		NULL	

4. Create the EMPLOYEES table with the following attributes:

(emp_id, emp_name, emp_salary, dept_no).

```
> CREATE TABLE EMPLOYEES (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    emp_salary INT,  
    dept_no INT,  
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)  
);
```

OUTPUT:

	Field	Type	Null	Key	Default	Extra
►	emp_id	int	NO	PRI	NULL	
	emp_name	varchar(50)	YES		NULL	
	emp_salary	int	YES		NULL	
	dept_no	int	YES	MUL	NULL	

DML Questions (Data Manipulation Language)

From TEACHER Table

1.Insert five records into the TEACHER table.

```
INSERT INTO TEACHER VALUES  
( 'Nakul Arora', 10, '2016-06-15', 'Mathematics', 'Delhi', 40000),  
( 'Vansh Tanwar', 20, '2017-07-10', 'Commerce', 'Mumbai', 45000),  
( 'Rahul Kumar', 10, '2015-05-20', 'Mathematics', 'Delhi', 42000),  
( 'Meena Patel', 30, '2018-08-25', 'Physics', 'Ahmedabad', 48000),  
( 'Kiran Das', 20, '2019-09-12', 'Commerce', 'Kolkata', 46000);|
```

OUTPUT:

	Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
▶	Nakul Arora	10	2016-06-15	Mathematics	Delhi	40000
	Vansh Tanwar	20	2017-07-10	Commerce	Mumbai	45000
	Rahul Kumar	10	2015-05-20	Mathematics	Delhi	42000
	Meena Patel	30	2018-08-25	Physics	Ahmedabad	48000
	Kiran Das	20	2019-09-12	Commerce	Kolkata	46000

2. Give an increment of 25% salary to teachers working in the Mathematics department.

```
SET SQL_SAFE_UPDATES = 0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
```

OUTPUT:

	Name	DeptName	Salary
▶	Nakul Arora	Mathematics	50000
	Rahul Kumar	Mathematics	52500

3. Give an increment of 15% salary to teachers working in the Commerce department.

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

OUTPUT:

	Name	DeptName	Salary
▶	Sunita Rao	Commerce	51750
	Kiran Das	Commerce	52900

From SALES Table

1.Insert five records into the SALES table.

```
INSERT INTO SALES VALUES
(201, 'Amit', 'Delhi', 120000, '1996-12-21'),
(202, 'Neha', 'Mumbai', 95000, '1995-07-14'),
(203, 'Rohit', 'Bangalore', 150000, '1994-12-09'),
(204, 'Sneha', 'Delhi', 87000, '1997-03-18'),
(205, 'Karan', 'Mumbai', 110000, '1993-12-25');
```

OUTPUT:

	Sales_No	Salesname	Branch	Salesamount	DOB
►	201	Amit	Delhi	120000	1996-12-21
	202	Neha	Mumbai	95000	1995-07-14
	203	Rohit	Bangalore	150000	1994-12-09
	204	Sneha	Delhi	87000	1997-03-18
	205	Karan	Mumbai	110000	1993-12-25

2.Calculate the total sales amount in each branch.

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT:

	Branch	Total_Sales
►	Delhi	207000
	Mumbai	205000
	Bangalore	150000

3.Calculate the average sales amount in each branch.

```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT:

	Branch	Average_Sales
▶	Delhi	103500.0000
	Mumbai	102500.0000
	Bangalore	150000.0000

4.Display the Salesname and DOB of salesmen born in the month of December with date format like 21-Dec-09.

```
SELECT Salesname,
       DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

OUTPUT:

	Salesname	DOB
▶	Amit	21-Dec-96
	Rohit	09-Dec-94
	Karan	25-Dec-93

5.Display the Salesname and DOB of salesmen in alphabetical order of the month.

```

SELECT Salesname,
       DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
ORDER BY MONTHNAME(DOB);

```

OUTPUT:

	Salesname	DOB
▶	Amit	21-Dec-96
	Rohit	09-Dec-94
	Karan	25-Dec-93
	Neha	14-Jul-95
	Sneha	18-Mar-97

From JOIN Operations

1.Display employee details and department details where the department numbers are the same in both tables.

```

SELECT e.emp_name, d.dept_name
FROM EMPLOYEES e
LEFT JOIN DEPARTMENTS d
ON e.dept_no = d.dept_no;

```

OUTPUT:

	emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
▶	1	Amit	50000	10	10	HR	Delhi
	3	Rahul	45000	10	10	HR	Delhi
	2	Neha	60000	20	20	IT	Bangalore
	4	Sneha	70000	20	20	IT	Bangalore
	5	Karan	40000	30	30	Sales	Mumbai

2.Display the employee name and department name using a LEFT OUTER JOIN.

```

SELECT e.emp_name, d.dept_name
FROM EMPLOYEES e
LEFT JOIN DEPARTMENTS d
ON e.dept_no = d.dept_no;

```

OUTPUT:

	emp_name	dept_name
▶	Amit	HR
	Neha	IT
	Rahul	HR
	Sneha	IT
	Karan	Sales

3.Display the employee name and department name using a RIGHT OUTER JOIN.

```

SELECT e.emp_name, d.dept_name
FROM EMPLOYEES e
RIGHT JOIN DEPARTMENTS d
ON e.dept_no = d.dept_no;

```

OUTPUT:

	emp_name	dept_name
▶	Amit	HR
	Rahul	HR
	Neha	IT
	Sneha	IT
	Karan	Sales

4.Display the details of employees who draw a salary greater than the average salary.


```
SELECT emp_id, emp_name, emp_salary
FROM EMPLOYEES
WHERE emp_salary >
      (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

OUTPUT:

	emp_id	emp_name	emp_salary
▶	2	Neha	60000
	4	Sneha	70000
	6	Vikas	55000
•	NULL	NULL	NULL

TCL Questions (Transaction Control Language)

1.Perform the ROLLBACK command on the TEACHER table.

```
START TRANSACTION;
UPDATE TEACHER
SET Salary = Salary + 2000;
ROLLBACK;
```

OUTPUT:

	Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
▶	Anil Kumar	10	2016-06-15	Mathematics	Delhi	40000
	Sunita Rao	20	2017-07-10	Commerce	Mumbai	45000
	Ravi Sharma	10	2015-05-20	Mathematics	Delhi	42000
	Meena Patel	30	2018-08-25	Physics	Ahmedabad	48000
	Kiran Das	20	2019-09-12	Commerce	Kolkata	46000

2.Perform the COMMIT command on the TEACHER table.

```

START TRANSACTION;
UPDATE TEACHER
SET Salary = Salary + 3000;
COMMIT;

```

OUTPUT:

	Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
►	Anil Kumar	10	2016-06-15	Mathematics	Delhi	43000
	Sunita Rao	20	2017-07-10	Commerce	Mumbai	48000
	Ravi Sharma	10	2015-05-20	Mathematics	Delhi	45000
	Meena Patel	30	2018-08-25	Physics	Ahmedabad	51000
	Kiran Das	20	2019-09-12	Commerce	Kolkata	49000

3. Write a query to implement a SAVEPOINT.

```

START TRANSACTION;
UPDATE TEACHER
SET Salary = Salary + 1000;
SAVEPOINT A;
UPDATE TEACHER
SET Salary = Salary + 500;
ROLLBACK TO A;
COMMIT;

```

OUTPUT:

	Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
►	Anil Kumar	10	2016-06-15	Mathematics	Delhi	44000
	Sunita Rao	20	2017-07-10	Commerce	Mumbai	49000
	Ravi Sharma	10	2015-05-20	Mathematics	Delhi	46000
	Meena Patel	30	2018-08-25	Physics	Ahmedabad	52000
	Kiran Das	20	2019-09-12	Commerce	Kolkata	50000

DCL Questions (Data Control Language)

1. Write a query to grant all privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT ALL  
ON EMPLOYEES  
TO DEPARTMENTS;
```

2. Write a query to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT SELECT, INSERT  
ON EMPLOYEES  
TO DEPARTMENTS;
```

3. Write a query to revoke all privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE ALL  
ON EMPLOYEES  
FROM DEPARTMENTS;
```

4. Write a query to revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE SELECT, INSERT  
ON EMPLOYEES  
FROM DEPARTMENTS;
```